

| STAT214 Chapter 8 An Introduction to Prediction Problems

| 1 Prediction Problems

| 1.1 Data-driven Predictions

Based on algorithms that quantify the relationships between a **response variable** that we are interested in predicting and the relevant **predictor variables** (often called the **predictors**, **predictive features**, **attributes**, or **covariates**)

| 1.2 Predictive Algorithms

- quantify the relationship between a response variable of interest and the predictive features using existing/observed data
- work by combining the values of the predictive features in a way such that the resulting combination—the **predicted** response—resembles the observed response variable as closely as possible.

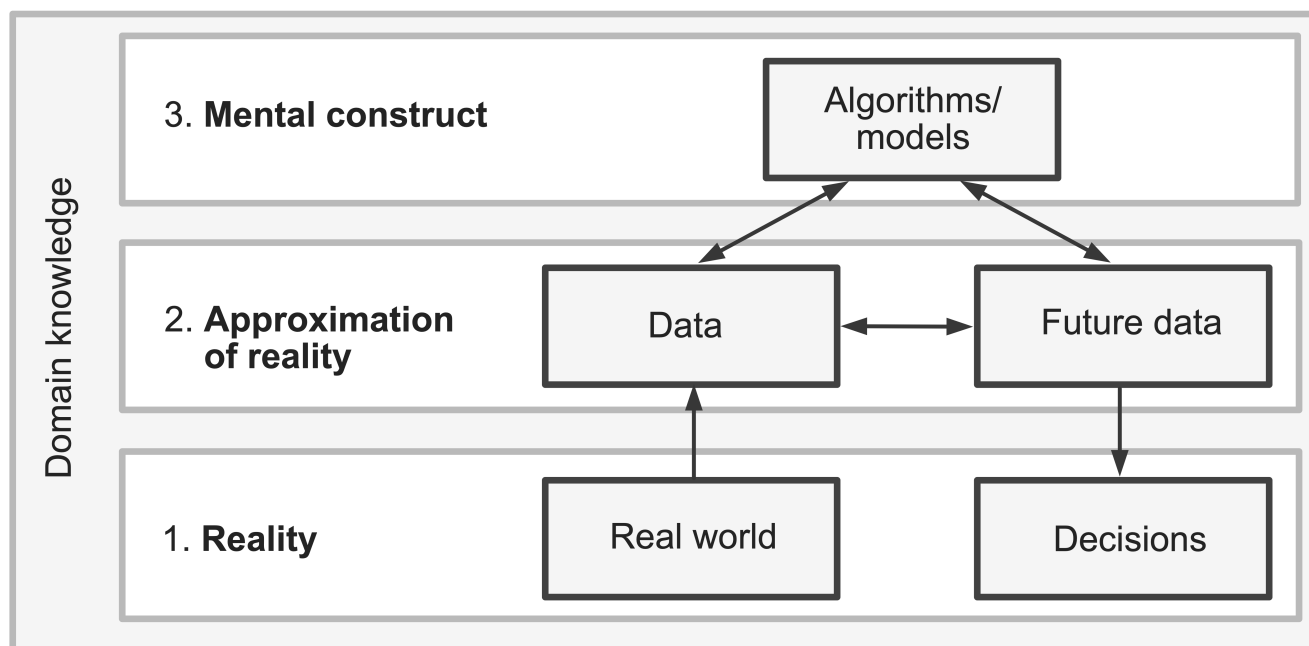
| 1.3 Two Main Types of ML Problems

- **Unsupervised learning**: Summarizing the relationships in the data when there is no response variable.
- **Supervised learning**: Summarizing the relationships in the data with a focus on a response variable of interest whose value is observed in current/past data but is not known for future data.

| 2 Connecting the Past, Present, and Future for Prediction Problems

| 2.1 Three Realms

- our data represents an **approximation of reality** (rather than reality itself)
- the algorithms (predictive or otherwise) that we train are just **mental constructs** that we apply to our data
- we need to pay particular attention to the link between the current and future data



| 2.2 Similarity between Future and Present Reality

- The assumption that the relevant underlying patterns and trends remain the same over time is major
- it is so important to evaluate our algorithms on withheld data that is a good representation of the future data to which they will be applied

- it is a good idea to compare the future data to the current data and to use domain knowledge to identify whether the underlying patterns, trends, and relationships have changed over time
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| 3 Setting up a Prediction Problem

| 3.1 Defining a Response Variable

- The first step in getting your prediction project off the ground is to translate your domain problem goal into a prediction problem.
- This involves precisely specifying what you want to predict

| 3.2 Labeled Data

- To train and evaluate a predictive algorithm, you need to have access to labeled data whose response is known
- The future data to which you will be applying the predictive algorithm will not have known response values

| 3.3 Two Most Common Types of Response Variables

- **Binary responses:** Binary response prediction problems are often called **classification** problems in traditional statistics and ML
- **Continuous responses:** Continuous response prediction problems are often called **regression problems** in traditional statistics

| 3.4 Defining Predictor Variables

- Ensure that the data contains appropriate predictive features
- For a variable to be an appropriate predictive feature, its value needs to be available (measured/observed) even when the response variable is not, and it should not depend on the response variable itself
- Row identifier variables or ID variables are also not appropriate predictive features

| 3.5 Quantity versus Quality

- We recommend focusing on quality over quantity
 - You will generally be better off having a smaller but more diverse dataset with a few highly relevant features than a larger but more uniform dataset with a large number of only vaguely relevant features
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| 4 PCS and Evaluating Prediction Algorithms

| 4.1 Predictability

- The most common way to demonstrate the *predictability* of the results of a predictive algorithm is to show that the algorithm generates accurate response predictions for data that *resembles the future data that the algorithm will be applied to in practice*.
- the data that you use to evaluate your predictions need to be **labeled**

| 4.1.1 Training, Validation, and Test Datasets

- When we don't have access to additional future or external *labeled* data similar to that to which we will be applying our algorithm, we will often instead withhold some validation data from our current observed labeled data to use for evaluation.
- your data should be split into training, validation, and test datasets in such a way that the validation and test datasets reflect the future, new, or external data to which you intend to apply your results, such as using a time-based, group-based, or random split

| 4.1.2 Reality Checks for Prediction Problems

- Additional evidence of predictability can be obtained from conducting *reality checks*

- to avoid confirmation bias, when presenting such results to domain experts, it is a good idea to show them multiple possible versions of the captured relationships, only one of which will be the actual version produced by your algorithm.

| 4.2 Stability

The stability of your response predictions can be assessed by training multiple versions of the predictive algorithm under different data and judgment call perturbations, and investigating the extent to which the predictions and predictive performance change.